

Point-in-Time Multi-Agent LLM Trading: A Reproducible LangGraph Stack with Regime-Conditioned Evaluation and Fair Backtesting

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Abstract

Large language models are increasingly used in financial decision pipelines, yet published agent stacks are often hard to reproduce and rarely evaluated under distinct market conditions on historical as-of dates. We present **Hindsight 20/20**, an open **LangGraph** stack with structured **Pydantic** outputs, point-in-time data access, optional **ticker anonymization**, and a fee-aware **paper backtest**. We evaluate the **full** pipeline on **RELIANCE.NS** and **TCS.NS** in **exogenous bear and bull windows** (per-ticker underlying return at most -10% or at least +10%), each from \$100,000 paper cash. In **both bear windows**, the agent loses less than buy-and-hold; in **bull windows**, outcomes diverge (RELIANCE lags B&H; TCS nearly tracks it). Frozen CSV artifacts reproduce tables and figures.

Keywords: algorithmic trading; large language models; multi-agent systems; reproducibility; point-in-time data; market regimes.

0.1 Introduction

Researchers are experimenting with LLMs inside trading workflows, but rarely report how the **same stack** behaves across **different underlying trends** or **more than one issuer**. **Hindsight 20/20** extends the open-source **TradingAgents** multi-agent LangGraph framework (Xiao et al., 2024) with point-in-time data access, optional ticker anonymization, and a reproducible backtest harness; we evaluate this extended stack under exogenous bear and bull windows. We evaluate the **full** pipeline on **RELIANCE.NS** and **TCS.NS** in four independent runs (bear and bull per ticker). Section 2 surveys related work; Section 3 the system; Section 4 results; Section 5 concludes; Section 6 lists contributions.

0.2 Related Work

LLM trading agents motivate multi-role graphs (Xiao et al., 2024; Yu et al., 2023; Yang et al., 2023). We emphasize reproducible engineering: one codepath, schema-constrained outputs, temporal controls, CSV artifacts.

Point-in-time data and leakage require explicit simulation horizons and optional anonymization.

Regime-split evaluation on the same implementation is uncommon; we report four frozen runs with **per-ticker calendars**, not regime prediction.

Execution realism via a paper ledger with stated fees.

0.3 Method / System

TradingAgentsGraph composes analysts, bull/bear **debate roles** (internal agents, distinct from market regimes), trader, and risk. Core analyst $\rightarrow$ debate $\rightarrow$ trader $\rightarrow$ risk routing follows **TradingAgents** (Xiao et al., 2024); our modifications focus on temporal data policy, schema-constrained outputs, and the backtest/evaluation pipeline described

below. Section 4 uses `PAPER_ABLATION=full`. `backtest_mvp.py --dates-csv` produces wide schedule CSVs.

0.3.1 Ticker anonymization

Large language models trained on public text often encode firm-specific narratives—brand reputation, sector themes, and historical events—tied to ticker symbols and company names. In a point-in-time backtest, that **pretraining leakage** can bias decisions: the same vendor data may elicit different agent behavior on `RELIANCE.NS` versus an unseen alias. Hind-sight 20/20 optionally **anonymizes the issuer** for all LLM-facing prompts and tool outputs: each real symbol maps to a deterministic synthetic id (`STOCK_####`, hash-derived); tool wrappers **de-anonymize** before market-data API calls, then **re-scrub** responses; news headlines additionally pass through a lightweight proper-noun scrubber. Frozen E1 runs use `enable_anonymization=True` (default in `backtest_mvp.py`). We do not ablate anonymization in Section 4—reporting with it **on** is a conservative choice that reduces name-based bias when comparing regimes and tickers.

0.3.2 Backtest metrics

The paper ledger records daily agent equity after fees. We report **total return** on agent equity; **maximum drawdown (MDD)** as peak-to-trough decline on the agent equity curve, with **buy-and-hold MDD** on the same underlying adjusted closes for comparison; and **Sharpe ratio** as mean daily simple return divided by its standard deviation, scaled by $\sqrt{252}$ ¹. **No risk-free rate is subtracted** in Frozen E1 (excess Sharpe is future work). Short windows (67–111 trading days) yield noisy Sharpe estimates for discrete-signal agents.

¹Annualization assumes 252 trading days per year (US equity convention). NSE calendars are typically ≈ 240 –250 trading days; we retain 252 for comparability with standard reported Sharpe ratios.

0.4 Experiments

0.4.1 Regime definition

For each ticker, calendar windows in **regime-timeranges** are labeled **bear** if underlying buy-and-hold return is at most **-10%**, **bull** if at least **+10%** over the processed window. Each run starts from **\$100,000** fresh cash. Windows differ by ticker.

0.4.2 Protocol and results

Table 1. Four frozen runs. Sharpe is computed on **agent** daily equity returns (annualized; see footnote in Section 3.4; **no risk-free rate** subtracted).

Ticker	Regime	Window	Days	B&H	Agent	B&H MDD	Ag. MDD	Sharpe	Buy/Hold/Sell
RELIANCE	Bear	2024-08-01 - 2025-01-03	111	-17.4%	-7.5%	21.0%	5.1%	-3.12	7/28/76
RELIANCE	Bull	2025-01-01 - 2025-06-03	108	+15.1%	+5.6%	11.0%	2.8%	-3.35	9/25/74
TCS	Bear	2024-12-02 - 2025-04-03	88	-20.4%	-4.8%	23.9%	6.8%	-1.83	12/24/52
TCS	Bull	2024-06-03 - 2024-09-03	67	+21.9%	+19.6%	5.5%	1.7%	+4.25	13/14/40

Bear (2/2): agent loses less than buy-and-hold on **total return** and shows **lower path drawdown** (e.g., RELIANCE bear: agent MDD 5.1% vs B&H MDD 21.0%), consistent with SELL-heavy signals and reduced long exposure. **Bull:** RELIANCE lags B&H; TCS nearly tracks B&H. **Sharpe** is negative in three of four runs—including RELIANCE bull (+5.6% return)—because daily equity volatility dominates on short windows; only TCS bull is Sharpe-positive (+4.25). Interpret Sharpe alongside total return and MDD, not in isolation.

0.4.3 Figures

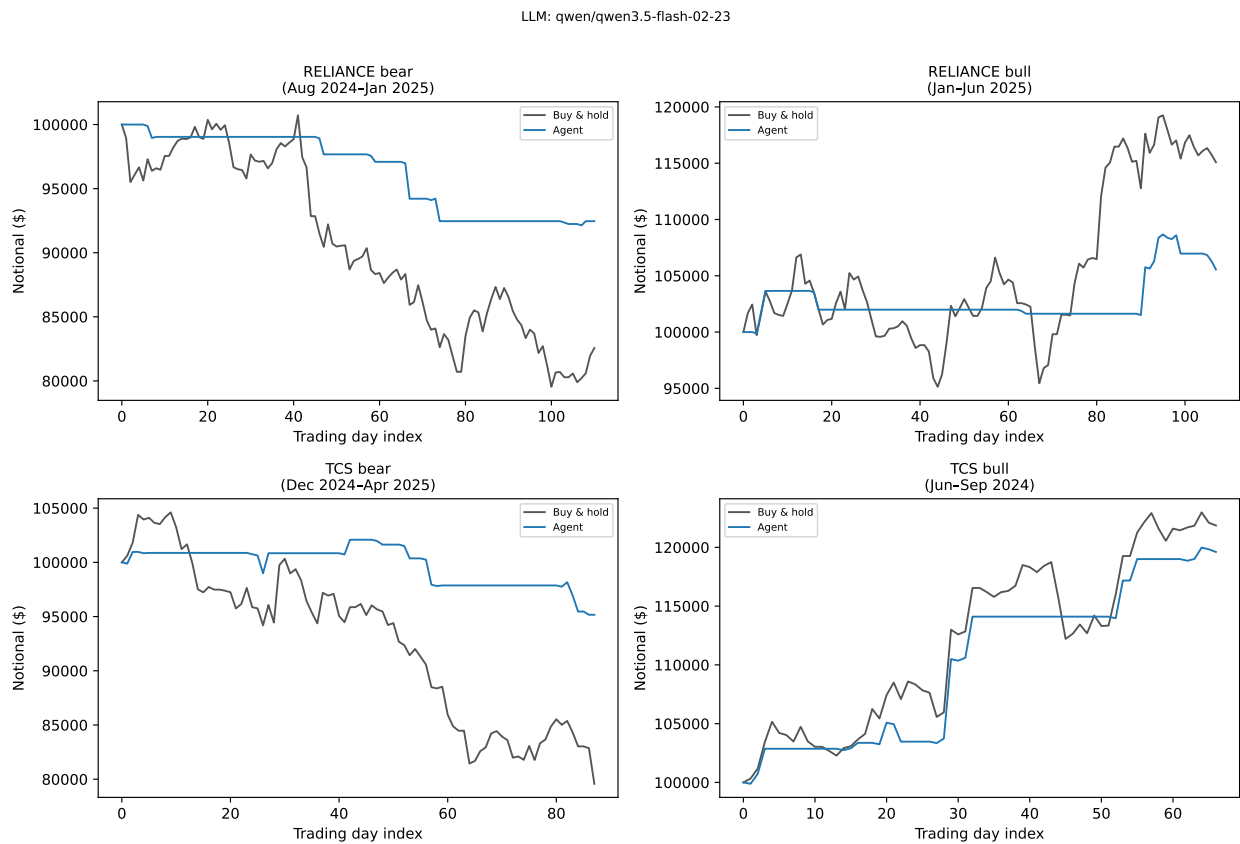


Figure 1: Figure 1. Agent vs buy-and-hold equity (2x2 grid).

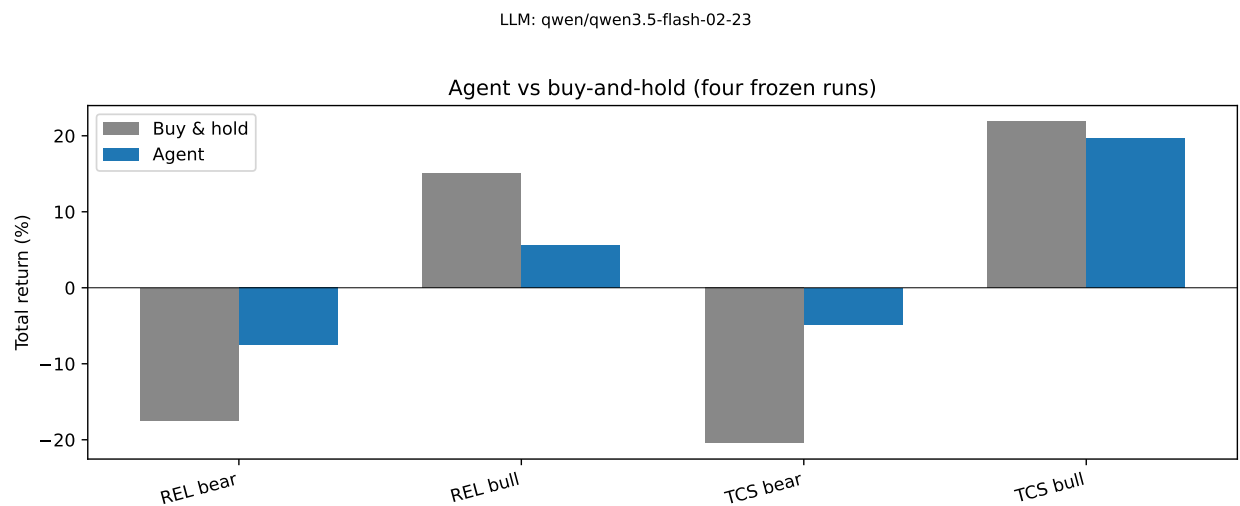


Figure 2: Figure 2. Total returns (four runs).

LLM: qwen/qwen3.5-flash-02-23

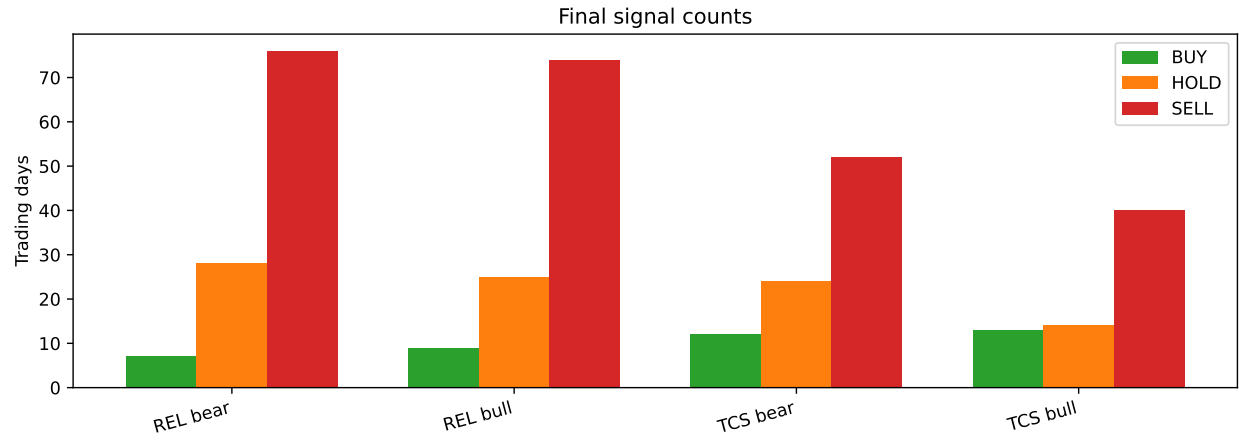


Figure 3: Figure 3. Signal counts.

LLM: qwen/qwen3.5-flash-02-23

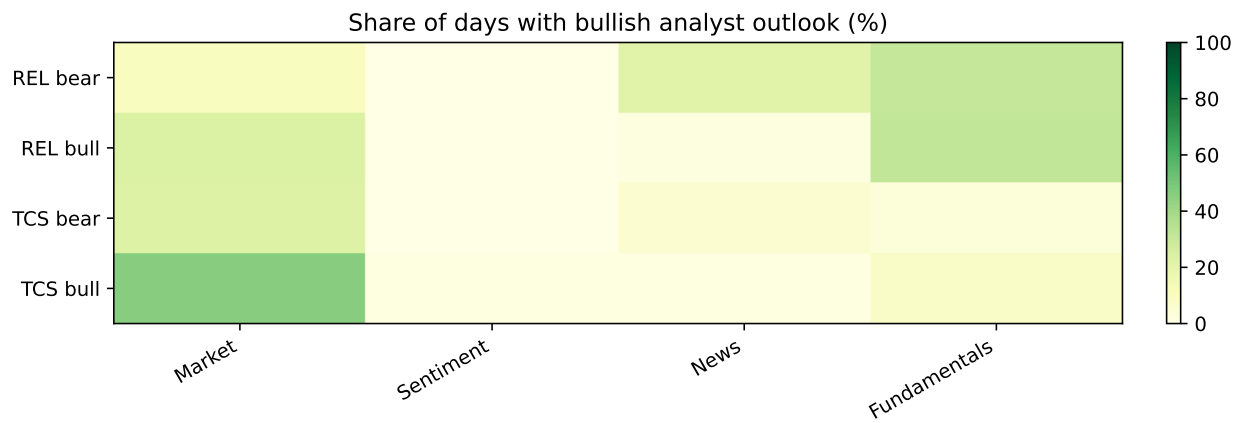


Figure 4: Figure 4. Bullish analyst-outlook share.

0.5 Conclusion

We presented an open LangGraph stack and **multi-ticker regime-conditioned** evaluation. Bear windows show relative downside protection on both **return and drawdown**; bull outcomes are heterogeneous. **Limitations:** two tickers, per-ticker calendars, one LLM; anonymization enabled but not ablated; Sharpe without a risk-free benchmark and on short windows.

0.6 Contributions

1. Multi-agent decision graph.
2. Multi-ticker regime-conditioned evaluation (four frozen CSVs).
3. Structured stage outputs.
4. Point-in-time policy and anonymization.
5. Backtest harness and analysis artifacts.

Acknowledgements

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